



Experiment 2

AIM: Setup Git and explore commands related to Version Control System (VCS)

Step 1: Download from <https://git-scm.com>

Step 2: Configuration

```
git config --global user.name "Kishan130"
```

```
git config --global user.email "kishanpatel041199@gmail.com"
```

Step 3:

```
C:\Users\kisha>git --version
```

```
git version 2.43.0.windows.1
```

```
C:\Users\kisha>git init
```

Reinitialized existing Git repository in C:/Users/kisha/.git/

```
C:\Users\kisha>mkdir ost
```

```
C:\Users\kisha>mkdir ost
```

A subdirectory or file ost already exists.

```
C:\Users\kisha>git init
```

Reinitialized existing Git repository in C:/Users/kisha/.git/

```
C:\Users\kisha>cd ost
```

```
C:\Users\kisha\ost>git init
```

Initialized empty Git repository in C:/Users/kisha/ost/.git/



```
C:\Users\kisha\ost>git status
```

```
On branch master
```

```
No commits yet
```

```
nothing to commit (create/copy files and use "git add" to track)
```

```
C:\Users\kisha\ost>echo sample.txt
```

```
sample.txt
```

```
C:\Users\kisha\ost>echo demo.py
```

```
demo.py
```

```
C:\Users\kisha\ost>git status
```

```
On branch master
```

```
No commits yet
```

```
nothing to commit (create/copy files and use "git add" to track)
```

```
C:\Users\kisha\ost>dir
```

```
Volume in drive C is OS
```

```
Volume Serial Number is E2F6-8AB0
```

```
Directory of C:\Users\kisha\ost
```

```
10-Feb-26 23:29 PM <DIR> .
```

```
10-Feb-26 23:22 PM <DIR> ..
```

```
0 File(s) 0 bytes
```

```
2 Dir(s) 17,472,352,256 bytes free
```

```
C:\Users\kisha\ost>echo sample.txt
```

```
sample.txt
```



```
C:\Users\kisha\ost>dir
```

```
Volume in drive C is OS
```

```
Volume Serial Number is E2F6-8AB0
```

```
Directory of C:\Users\kisha\ost
```

```
10-Feb-26 23:29 PM <DIR> .
```

```
10-Feb-26 23:22 PM <DIR> ..
```

```
0 File(s) 0 bytes
```

```
2 Dir(s) 17,472,425,984 bytes free
```

```
C:\Users\kisha\ost>echo Hello> sample.txt
```

```
C:\Users\kisha\ost>dir
```

```
Volume in drive C is OS
```

```
Volume Serial Number is E2F6-8AB0
```

```
Directory of C:\Users\kisha\ost
```

```
10-Feb-26 23:35 PM <DIR> .
```

```
10-Feb-26 23:22 PM <DIR> ..
```

```
10-Feb-26 23:35 PM 7 sample.txt
```

```
1 File(s) 7 bytes
```

```
2 Dir(s) 17,471,066,112 bytes free
```

```
C:\Users\kisha\ost>type sample.txt
```

```
Hello
```



```
C:\Users\kisha\ost>git commit -m "Initial Commit"
```

On branch master

Initial commit

Untracked files:

(use "git add <file>..." to include in what will be committed)

sample.txt

nothing added to commit but untracked files present (use "git add" to track)

```
C:\Users\kisha\ost>git config --global init.defaultBranch main
```

```
C:\Users\kisha\ost>git commit -m "Initial Commit"
```

On branch master

Initial commit

Untracked files:

(use "git add <file>..." to include in what will be committed)

sample.txt

nothing added to commit but untracked files present (use "git add" to track)

```
C:\Users\kisha\ost>git add sample.txt
```

```
C:\Users\kisha\ost>git commit -m "Initial Commit"
```

```
[master (root-commit) df439d9] Initial Commit
```

```
1 file changed, 1 insertion(+)
```

```
create mode 100644 sample.txt
```

```
C:\Users\kisha\ost>git branch
```

```
* master
```



```
C:\Users\kisha\ost>git branch -m master main
```

```
C:\Users\kisha\ost>git branch
```

```
* main
```

```
C:\Users\kisha\ost>git checkout main
```

```
Already on 'main'
```

```
C:\Users\kisha\ost>git checkout -b dev
```

```
Switched to a new branch 'dev'
```

```
C:\Users\kisha\ost>echo Hello from dev>> sample.txt
```

```
C:\Users\kisha\ost>git add .
```

```
C:\Users\kisha\ost>git commit -m "Update in dev"
```

```
[dev de8fe00] Update in dev
```

```
1 file changed, 1 insertion(+)
```

```
C:\Users\kisha\ost>git checkout main
```

```
Switched to branch 'main'
```

```
C:\Users\kisha\ost>git merge dev
```

```
Updating df439d9..de8fe00
```

```
Fast-forward
```

```
sample.txt | 1 +
```

```
1 file changed, 1 insertion(+)
```



```
C:\Users\kisha\ost>dir
```

Volume in drive C has no label.

Volume Serial Number is FCE0-3C72

Directory of C:\Users\kisha\ost

```
10-02-2026 23:26 <DIR>      .
10-02-2026 23:18 <DIR>      ..
10-02-2026 23:26          228,913 01CE0618 - Open Source Technologies Syllabus.pdf
10-02-2026 23:26          19,168 Experiment-1-OST.docx
10-02-2026 23:26          90,156 Experiment-1-OST.pdf
10-02-2026 23:20              0 git
10-02-2026 23:26              21 README.md
10-02-2026 23:20              23 sample.txt
        6 File(s)      338,281 bytes
        2 Dir(s) 23,107,969,024 bytes free
```

```
C:\Users\kisha\ost>type sample.txt
```

Hello

Hello from dev

```
C:\Users\kisha\ost>git remote -v
```

```
C:\Users\kisha\ost>git remote add origin https://github.com/Kishan130/OST-Kishan-
Vachhani
```

error: remote origin already exists.

```
C:\Users\kisha\ost>git remote -v
```

```
origin https://github.com/Kishan130/OST-Kishan-Vachhani (fetch)
```

```
origin https://github.com/Kishan130/OST-Kishan-Vachhani (push)
```



```
C:\Users\kisha\ost>git push -u origin main
```

```
To https://github.com/Kishan130/OST-Kishan-Vachhani
```

```
! [rejected]    main -> main (non-fast-forward)
```

```
error: failed to push some refs to 'https://github.com/Kishan130/OST-Kishan-Vachhani'
```

```
hint: Updates were rejected because the tip of your current branch is behind
```

```
hint: its remote counterpart. If you want to integrate the remote changes,
```

```
hint: use 'git pull' before pushing again.
```

```
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

```
C:\Users\kisha\ost>
```

```
C:\Users\kisha\ost>git push -u origin main
```

```
Enumerating objects: 10, done.
```

```
Counting objects: 100% (10/10), done.
```

```
Delta compression using up to 12 threads
```

```
Compressing objects: 100% (5/5), done.
```

```
Writing objects: 100% (9/9), 793 bytes | 79.00 KiB/s, done.
```

```
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
```

```
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
```

```
To https://github.com/Kishan130/OST-Kishan-Vachhani
```

```
7b70dba..3b0d4ee main -> main
```

```
branch 'main' set up to track 'origin/main'.
```

Kishan130 / OST-Kishan-Vachhani

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

OST-Kishan-Vachhani

Public

Pin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

Add file

<> Code

Kishan130

Merge remote changes

3b0d4ee · 3 minutes ago

8 Commits

01CE0618 - Open Source Technologies Syllabu...	Add files via upload	3 weeks ago
Experiment-1-OST.docx	Rename Experiment1OST97.docx to Experiment-1-OST.docx	3 weeks ago
Experiment-1-OST.pdf	Add files via upload	3 weeks ago
README.md	Initial commit	3 weeks ago
git	Update in dev	15 minutes ago
sample.txt	Update in dev	15 minutes ago

README

OST-Kishan-Vachhani

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)


```
Command Prompt
C:\Users\kisha\ost>type sample.txt
Hello
Hello from dev

C:\Users\kisha\ost>git remote -v
origin https://github.com/Kishan130/OST-Kishan-Vachhani (fetch)
origin https://github.com/Kishan130/OST-Kishan-Vachhani (push)

C:\Users\kisha\ost>git remote add origin https://github.com/Kishan130/OST-Ki
shan-Vachhani
error: remote origin already exists.

C:\Users\kisha\ost>git push -u origin main
To https://github.com/Kishan130/OST-Kishan-Vachhani
! [rejected]        main -> main (non-fast-forward)
error: failed to push some refs to 'https://github.com/Kishan130/OST-Kishan-
Vachhani'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

C:\Users\kisha\ost>
C:\Users\kisha\ost>
C:\Users\kisha\ost>git pull origin main --allow-unrelated-histories
error: You have not concluded your merge (MERGE_HEAD exists).
hint: Please, commit your changes before merging.
fatal: Exiting because of unfinished merge.

C:\Users\kisha\ost>git commit -m "Merge remote changes"
[main 3b0d4ee] Merge remote changes

C:\Users\kisha\ost>git push -u origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (9/9), 793 bytes | 79.00 KiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Kishan130/OST-Kishan-Vachhani
7b70dba..3b0d4ee  main -> main
branch 'main' set up to track 'origin/main'.
```